

Knowledge Report: Artificial Intelligence

Summary

This domain is for use in documentation examples without needing permission. Learn more Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. It is a field of research in engineering, mathematics, and computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals.[1] High-profile applications of AI include advanced web search engines, chatbots, virtual assistants, autonomous vehicles, and play and analysis in strategy games (e.g., chess and Go). Since the 2020s, generative AI has become widely available to generate images, audio, and videos from text prompts. The traditional goals of AI research include learning, reasoning, knowledge representation, planning, natural language processing, and perception, as well as support for robotics.[a] To reach these goals, AI researchers use techniques including state space search and mathematical optimization, formal logic, artificial neural networks, and methods based on statistics, operations research, and economics.[b] AI also draws upon psychology, linguistics, philosophy, neuroscience, and other fields.[2] Some companies, such as OpenAI, Google DeepMind, and Meta, aim to create artificial general intelligence (AGI)AI that can complete nearly any cognitive task at least as well as a human.[3] Artificial intelligence was founded as an academic discipline in 1956.[4] The field went through multiple cycles of optimism throughout its history,[5][6] followed by periods of disappointment and loss of funding, known as AI winters.[7][8] Funding and interest increased substantially after 2012, when graphics processing units (GPUs) started being used to accelerate neural networks, and deep learning outperformed previous AI techniques.[9] This growth accelerated further after 2017 with the transformer architecture.[10] In the 2020s, an AI boom coincided with advances in generative AI, which allowed for the creation and modification of media.

Keywords

intelligence, artificial, learning, data, machine, models, deep, neural

Content

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multiple cycles of optimism throughout its history,[5][6] followed by periods of disappointment and loss of funding, known as AI winters.[7][8] Funding and interest increased substantially after 2012, when graphics processing units (GPUs) started being used to accelerate neural networks, and deep learning outperformed previous AI techniques.[9] This growth accelerated further after 2017 with the transformer architecture.[10] In the 2020s, an AI boom coincided with advances in generative AI, which allowed for the creation and modification of media. In addition to AI safety and unintended consequences and harms from the use of AI, ethical concerns, AI's long-term effects, environmental effects,[11] and potential existential risks have prompted discussions of AI regulation.

Goals The general problem of simulating (or creating) intelligence has been broken down into subproblems. These consist of specific traits or capabilities that researchers expect an intelligent system to display. The traits described below have received the most attention and cover the scope of AI research.

[a] Reasoning and problem-solving Early researchers developed algorithms that imitated step-by-step reasoning that humans use when solving puzzles or making logical deductions.[12] By the late 1980s and 1990s, methods were developed for dealing with uncertain or incomplete information, employing concepts from probability and economics.[13] Many of these algorithms were insufficient for solving large reasoning problems because they experienced a combinatorial explosion, meaning they become exponentially slower as the problems grow.[14] Even humans rarely use the step-by-step deduction that early AI research could model. Humans solve most of their problems using fast, intuitive judgments.[15] Reasoning models, a type of large language model (LLM) trained to generate intermediate chains-of-thought, emerged in 2024 and allowed improved performance on complex problems in mathematics and coding.[16] These models can produce incorrect outputs or hallucinations, unlike symbolic reasoning systems.[17]

Knowledge representation Knowledge representation and knowledge engineering[18] allow AI programs to answer questions intelligently and make deductions about real-world facts. Formal knowledge representations are used in content-based indexing and retrieval,[19] scene interpretation,[20] clinical decision support,[21] knowledge discovery (mining interesting and actionable inferences from large databases),[22] and other areas.[23] A knowledge base is a body of knowledge represented in a form that can be used by a program. An ontology is the set of objects, relations, concepts, and properties used by a particular domain of knowledge.[24] Knowledge bases need to represent things such as objects, properties, categories, and relations between objects;[25] situations, events, states, and time;[26] causes and effects;[27] knowledge about knowledge (what we know about what other people know);[28] default reasoning (things humans assume are true until they are told differently and will remain true even when other facts are changing);[29] and many other aspects and domains of knowledge. Among the most difficult problems in knowledge representation are the breadth of commonsense knowledge (the set of atomic facts the average person knows is enormous)[30] and the sub-symbolic form of most commonsense knowledge (much of what people know is not represented as facts or statements they can express verbally).[15]

There is also the difficulty of knowledge acquisition, the problem of obtaining knowledge for AI applications.

[c] Planning and decision-making An agent is any entity (artificial or not) that perceives and takes actions in the world. A rational agent has goals or preferences and takes actions to make them happen.[d][33] In automated planning, the agent has a specific goal.[34] In automated decision-making, the agent has preferences; there are some situations it would prefer to be in, and some situations it is trying to avoid. The decision-making agent assigns a number to each situation (called its utility) that measures how much the agent prefers it. For each possible action, it can calculate the expected utility: the utility of all possible outcomes of the action, weighted by the probability that the outcome will occur. It can then choose the action with the maximum expected utility.[35] In classical planning, the agent knows exactly what the effect of any action will be.[36] In most real-world problems, however, the agent may not be certain about the situation it is in (it is unknown or unobservable) and it may not know for certain what will happen after each possible action (it is not deterministic). It must choose an action by making a probabilistic guess and then reassess the situation to see if the action worked.[37]

Alongside thorough testing and improvement based on previous decisions, having an explanation for why the agent took certain decisions is a way to build trust, especially when the decisions have to be relied upon.[38] In some problems, the agent's preferences may be uncertain, especially if there are other agents or humans involved. These preferences may be learned (e.g., with inverse reinforcement learning), or the agent can seek information to improve them.[39] Information value theory can be used to weigh the value of exploratory or experimental actions.[40] The space of

possible future actions and situations is typically intractably large, so the agents must take actions and evaluate situations while being uncertain of the outcome. A Markov decision process has a transition model that describes the probability that a particular action will change the state in a particular way and a reward function that supplies the utility of each state and the cost of each action. A policy associates a decision with each possible state. The policy could be calculated (e.g., by policy iteration), determined by a heuristic, or learned.[41] Game theory describes the rational behavior of multiple interacting agents and is used in AI programs that make decisions involving other agents.[42] Learning Machine learning is the study of programs that can improve their performance on a given task automatically.[43] It has been a part of AI from the beginning.[e] There are several kinds of machine learning: - Unsupervised learning analyzes a stream of data, finds patterns, and makes predictions without any other guidance.[46] - Supervised learning requires labeling the training data with the expected answers, and comes in two main varieties: classification (where the program must learn to predict what category the input belongs in) and regression (where the program must deduce a numeric function based on numeric input).[47] - Reinforcement learning is when the agent is rewarded for good responses and punished for bad ones. The agent learns to choose responses that are classified as good.[48] - Transfer learning is when the knowledge gained from one problem is applied to a new problem.[49] - Deep learning is a type of machine learning that runs inputs through biologically inspired artificial neural networks for all of these types of learning.[50] Computational learning theory can assess learners by computational complexity, sample complexity (how much data is required), or other notions of optimization.[51] Natural language processing Natural language processing (NLP) allows programs to read, write, and communicate in human languages.[52] Specific problems include speech recognition, speech synthesis, machine translation, information extraction, information retrieval, and question answering.[53] Early work, based on Noam Chomsky's generative grammar and semantic networks, had difficulty with word-sense disambiguation[f] unless restricted to small domains called micro-worlds (due to the common sense knowledge problem[30]). British linguist and philosopher Margaret Masterman believed it was meaning and not grammar that was the key to understanding languages,[54] and that dictionaries and especially thesauri should be the basis of computational language structure.[55] Modern deep learning techniques for NLP include word embedding (representing words, typically as vectors encoding their meaning),[56] transformers (